Next-Day Rain Classification for the Colombo Area

IT5514 - Applied Machine Learning
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1. Problem definition and dataset

The task is to classify the following day as wet or dry in the Colombo area using recent weather and the time of year. A wet day means at least 1 mm of precipitation. Rain affects transport and outdoor work, but this target does not describe rainfall severity or flood risk.

I compiled the dataset through the Open-Meteo Historical Weather API, selecting ECMWF ERA5 [1]. The saved extract contains 4,383 daily records from 1 January 2014 to 31 December 2025, aggregated in Asia/Colombo and downloaded on 27 October 2026. The requested point was 6.9271 N, 79.8612 E; the returned grid location was 7.0 N, 80.0 E. These are therefore gridded estimates for the Colombo area, rather than station measurements.

Feature group	**Description**
Preceding-day rain	Daily precipitation (mm)
Humidity	Daily mean relative humidity (%)
Maximum/minimum temperature	Daily air-temperature extremes (C)
Wind	Daily maximum 10 m wind speed (km/h)
Pressure	Daily mean surface pressure (hPa)
3-day and 7-day mean rain	Trailing means ending on the preceding day (mm/day)
Pressure change	Preceding-day pressure minus the day before (hPa)
Season sine and cosine	Two values calculated from the target date

Preprocessing and ethical use

Dates were unique and consecutive, with no missing raw values. Rainfall, humidity, wind and temperature consistency were checked. No imputation or outlier removal was needed. Removing the first six incomplete history windows and the final unavailable next-day target left 4,376 rows and 11 numerical predictors. Weather-history inputs use only the completed day before the target and earlier days. No categorical encoding was required.

The source is public and contains no personal or sensitive data. Open-Meteo and ECMWF ERA5 are credited under CC BY 4.0 [2]. The raw response, daily CSV, source request, units and checksum accompany the code.

2. Algorithm selection and training

Regularised Quadratic Discriminant Analysis (QDA) was selected because it is outside the supplied lecture-algorithm list and uses neither deep learning nor image processing. QDA fits a multivariate Gaussian density to each class, with separate means and covariance matrices, then applies Bayes' rule [3]. Unequal covariances give a quadratic decision boundary and interactions between weather variables. Regularisation reduces instability when rainfall-history predictors are correlated.

Logistic regression fits linear log odds directly; QDA models the inputs within each class. A decision tree divides the input space using rules, while QDA uses continuous class densities. KNN uses distances to stored examples; QDA summarises each class through fitted parameters. It offers a compact nonlinear model, but its Gaussian assumptions may fit rainfall poorly.

Chronological split and transformations

Partition	Target dates	Rows	Wet days
Train	2014-01-08 to 2021-12-31	2,915	2,116
Validation	2022-01-01 to 2023-12-31	730	572
Test	2024-01-01 to 2025-12-31	731	556

The split preserves time order. Next-day rainfall is excluded from the predictors; the known target date supplies sine and cosine of 2*pi*(day-of-year - 1)/365.25. The three rain predictors receive log(1 + rainfall). Standardisation and class priors use training rows only, with no resampling or refitting on validation/test data.

Hyperparameter selection

Covariance shrinkage uses (1 - alpha) times the fitted covariance plus alpha times the identity, in standardised coordinates. I compared eight alpha values using validation AUC. Alpha 0.75 gave the best value, 0.8978. Validation balanced accuracy then selected a wet-day cutoff of 0.78 from 0.20-0.80 in steps of 0.005. Test rows were not used for either choice.

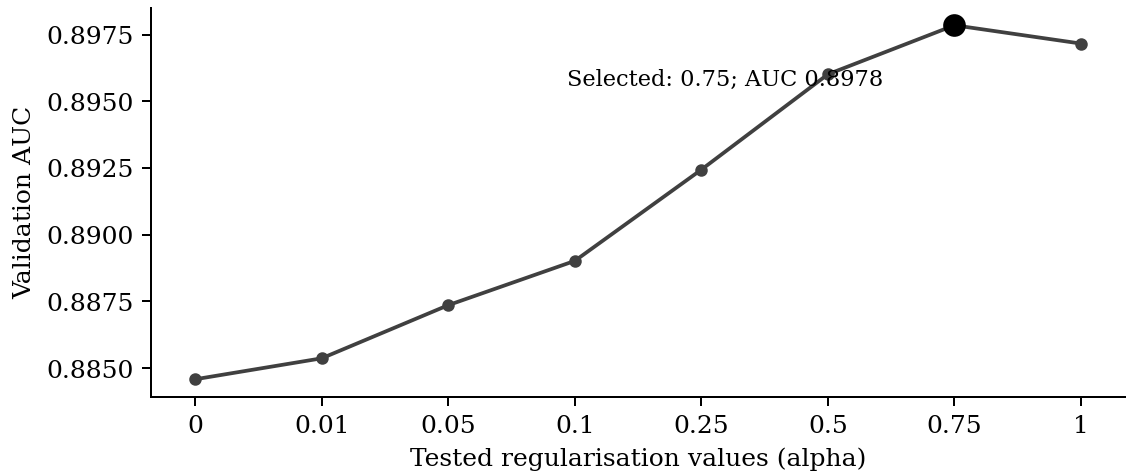


Figure 1. Validation AUC for the eight tested covariance-regularisation values.

Baselines use training-month wet frequencies, preceding-day wet/dry persistence and constant training prevalence. Logistic regression uses train-only scaling and C = 1. Its cutoff and the monthly-climatology cutoff are also selected on validation rows.

3. Evaluation results

AUC measures probability ranking. Accuracy counts correct decisions; balanced accuracy weights wet and dry recall equally. F1 combines wet-day precision and recall. Average precision (AP) summarises precision-recall ranking. Brier loss measures squared probability error; lower is better. These measures are used together because wet days are common.

Model	AUC	Accuracy	Balanced accuracy	F1	Brier loss
Regularised QDA	0.873	0.814	0.786	0.873	0.132
Logistic regression	0.885	0.807	0.801	0.865	0.107
Monthly climatology	0.825	0.811	0.807	0.868	0.131
Rain persistence	0.775	0.836	0.775	0.892	0.164
Constant train prevalence	0.500	0.761	0.500	0.864	0.183

Across 731 test days, QDA achieved AUC 0.873, accuracy 81.4%, balanced accuracy 0.786 and F1 0.873. Logistic regression gave higher AUC and lower Brier loss. Persistence gave higher accuracy and F1, while monthly climatology gave higher balanced accuracy. QDA did not outperform all simpler alternatives.

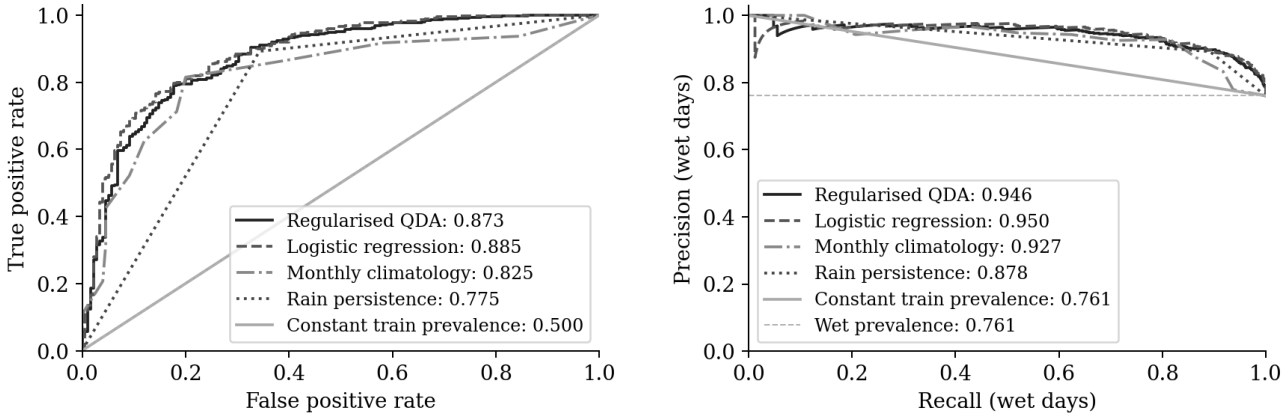


Figure 2. Test ROC (left) and precision-recall (right); legend values are AUC and AP respectively. The precision-recall reference is the 0.761 wet-day prevalence.

Errors at the fixed 0.78 cutoff

Actual / predicted	Dry	Wet
Dry	128	47
Wet	89	467

There were 89 missed wet days and 47 false wet alerts. Wet-day recall was 83.99%, dry-day recall 73.14% and wet-day precision 90.86%. Predicting wet every day would already give 76.06% accuracy. QDA's AP was 0.946, but its Brier loss of 0.132 and confident wrong predictions caution against treating its probabilities as calibrated forecasts.

4. Explainability and interpretation

Interventional SHAP explains QDA's raw log-odds score [4] against all 2,915 training rows. The quadratic score allows exact allocation of linear, squared and pairwise terms. The base plus all contributions reconstructs the score; a logistic transform gives the probability. Contributions are not percentage points.

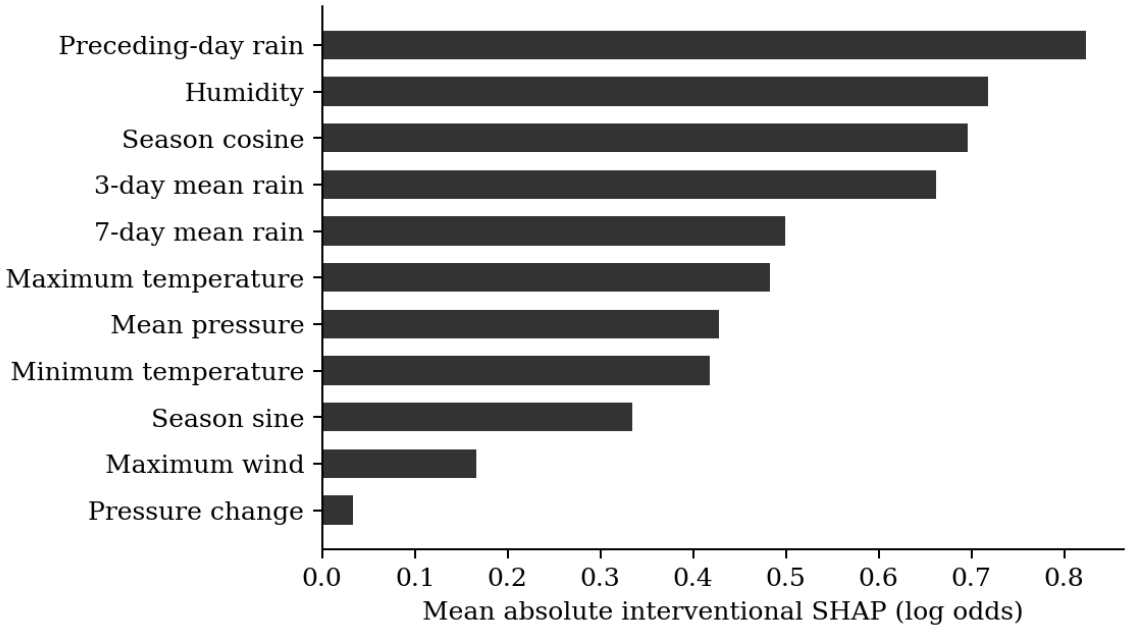


Figure 3. Mean absolute SHAP for 240 evenly spaced test days; training rows form the background.

Preceding-day rain contributed most on average (0.823 log odds), followed by humidity (0.718) and cosine season (0.696). Their prominence is consistent with recent rain, moisture and seasonal weather patterns, but does not establish causation.

Individual predictions

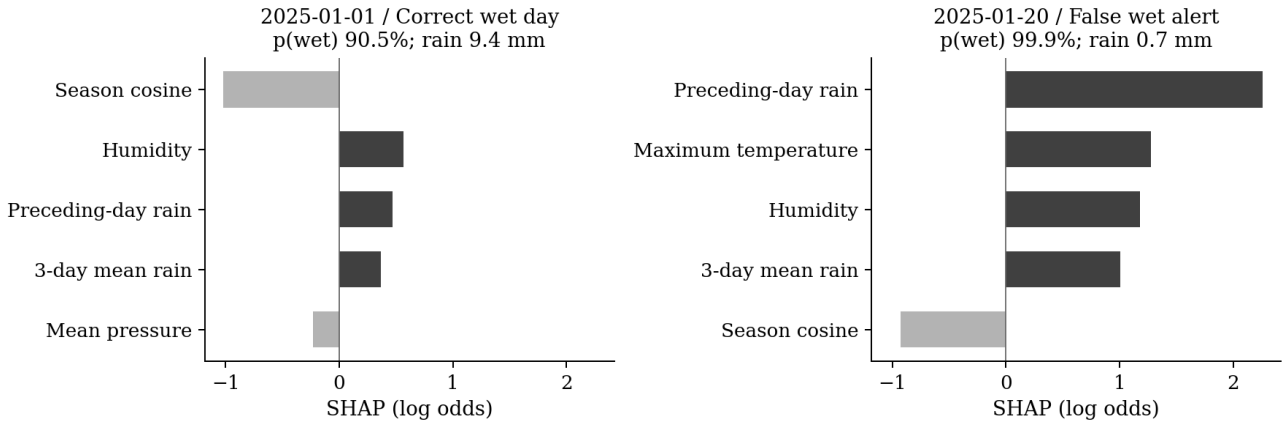


Figure 4. Five largest contributions for two test examples. Dark bars increase wet-day log odds; light bars reduce them. The frontend shows all eleven.

For 1 January 2025, humidity contributed +0.568 log odds and cosine season -1.020. The base 2.273 plus all contributions gave 2.257, or 90.5% wet probability. The recorded 9.4 mm confirmed a wet day.

The false alert on 20 January received 99.9% probability although only 0.7 mm fell. SHAP explains the fitted model, including its errors. Related rain windows and the seasonal coordinates can share credit. Interventional substitutions can break these relationships.

5. Critical discussion

Data availability is the main limitation. ERA5 is retrospective reanalysis, so complete inputs would not necessarily have been available when a next-day forecast was issued. Chronological splitting keeps later targets out of training, but does not establish operational forecast skill. The returned grid cell also differs from the requested point and misses local showers. Station measurements are needed to assess local accuracy.

QDA assumes Gaussian class densities. Rainfall has many small values and a long upper tail, which log transformation cannot fully correct. Rolling rain windows overlap and the season coordinates are dependent. Regularisation reduces instability without making these assumptions true. Logistic regression's stronger AUC and Brier result shows that greater model flexibility did not automatically improve generalisation.

Wet days dominate the data, hiding weaker dry-day performance in aggregate accuracy. A single-city model gives uneven geographic coverage and may be less useful for neighbourhoods or occupations exposed to different rainfall risks. The 1 mm boundary assigns small amounts such as 0.7 mm and 1.7 mm to different classes. Only two years were held out, and validation data were reused for alpha and cutoff selection. The reported differences therefore do not establish statistical superiority.

Probabilities were not independently calibrated. Confident wrong outputs could mislead people planning outdoor work; missed rain and false alerts also have different costs. The application is presented as a historical research demonstration, with no flood or safety advice. Calibration, wider geographic checks and inputs actually available at forecast time would be needed before practical use.

6. Frontend integration

The Flask and HTML/CSS frontend accepts nine weather-history inputs and a target date, then displays wet/dry classification, probability and an exact SHAP explanation for that entry. A recorded example can be edited and submitted again. Numerical limits, temperature consistency and rolling rain totals are checked before prediction. Edits remove the recorded-outcome label, and inputs outside the training range receive a warning.

Test results, six generated figures, source attribution and a CSV download are included. The saved QDA model is loaded at startup and does not train during requests. The application is prepared for Linux Azure App Service with Gunicorn.

References

[1] Open-Meteo. Historical Weather API documentation. ERA5 data source and variables.

[2] Creative Commons. Attribution 4.0 International licence.

[3] Scikit-learn. QuadraticDiscriminantAnalysis, version 1.8 documentation.

[4] Lundberg, S. M. and Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems, 30.